

Chemistry 1 (CYL1010)

Practice set 1

Q.1. A light bulb of 60 W emits at a wavelength of $0.5 \mu\text{m}$. Calculate the number of photons emitted per second.

Q.2. Calculate the energy in Joule and the wavenumber in cm^{-1} of:

(a) A photon with wavelength $1 \mu\text{m}$

(b) A photon with frequency $6.00 \times 10^{14} \text{ Hz}$.

Q.3. A non-degenerate excited level of a molecule lies 10 cm^{-1} above the non-degenerate ground level. Calculate the population of the excited level relative to that of the ground state at $T = 1 \text{ K}$ and $T = 300 \text{ K}$. What do you infer from the result?

Q.4. Check whether the following functions are acceptable or not? Provide reasons.

a) $f(x) = Ae^{-x^2}$

b) $f(x) = A \sin\left(\frac{1}{x}\right) \quad x \neq 0$

c) $f(x) = \frac{1}{(1+x^2)} A \quad x \neq 0$

Q.5. The wavefunction is

$$\psi(x) = A(1 + x), -1 \leq x \leq 1$$

a. Find the normalization constant A .

b. Calculate the probability of finding the particle in the region $-1 \leq x \leq 0$.

Q.6. The wave function for a certain particle is $\psi = A \cos^2(x)$ for $-\pi/2 \leq x \leq \pi/2$;

(a) Find the value of A .

(b) Find the probability that the particle be found between $x = 0$ and $x = \pi/4$.

Q.7. Determine whether the following operators are linear or not

(a) $Af(x) = \sqrt{f(x)}$

(c) $Af(x) = \ln f(x)$

(b) $Af(x) = x^2 f(x)$

(d) $Af(x) = \int_0^\infty e^{-sx} f(x) dx$

Q.8. If A and B are two operators such that $[A, B] = 1$ then Show that $[A, B^2] = 2B$

Q.9. Show that $[x, p_x] = i\hbar$

Q.10. The time independent wavefunction of a particle of mass m moving in a potential

$$V = \alpha^2 x^2 \text{ is } \psi(x) = \exp\left(-\sqrt{\frac{m\alpha^2}{2\hbar^2}} x^2\right), \alpha \text{ is a constant}$$

Find the total energy of the system.

Q.11. State True or False with justification:

- a) A wavefunction can be complex but must give real probabilities.
- b) A wavefunction must always be differentiable everywhere.
- c) The wavefunction itself has direct physical meaning.

Q.12. Find the action of the operators on the given function:

Given the wavefunction

$$\psi(x) = x^2 e^{-x}$$

- a) Find $\hat{x}\psi(x)$
- b) Find $\hat{p}\psi(x)$
- c) Find $\hat{p}^2\psi(x)$

Q.13. Check whether the function

$$\psi(x) = e^{ikx}$$

is an eigenfunction of:

- a) Momentum operator $\hat{p}$
- b) Energy operator $\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$

If yes, find the corresponding eigenvalues.

Q.14. For the normalized wavefunction

$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right), 0 \leq x \leq L$$

- a) Calculate $\langle x \rangle$
- b) Calculate $\langle p \rangle$
- c) Calculate $\langle p^2 \rangle$

Q.15. Given operators $\hat{A} = x$ and $\hat{B} = \frac{d}{dx}$:

- a) Find $\hat{A}\hat{B}\psi(x)$
- b) Find $\hat{B}\hat{A}\psi(x)$
- c) Evaluate $[\hat{A}, \hat{B}]$

Q.16. Prove that the K.E operator and the Hamiltonian operator are Hermitian operators.

Q.17. Evaluate $[x, p_x], [z, p_y]$

Q.18. The wave function of a particle of mass m moving in a potential V is given by $\psi(x, t) = A \exp\left[-ikt - \frac{km}{\hbar} x^2\right]$ where A and k are constants. Find the explicit form of the potential V .

Q.19. If ψ_1 and ψ_2 are two eigen functions of an operator $\hat{A}$ with eigen values E_1 and E_2 , then any linear combination of ψ_1 and ψ_2 will also be an eigen function of the operator $\hat{A}$ if

- (i) $E_1 = E_2$ (ii) $E_1 \neq E_2$
- (iii) If E_1 is a multiple of E_2 (iv) none of these

Q.20. A ball of mass 1g confined to a 1D box of length 0.1m moves with a velocity of 0.01 m/s. Calculate the quantum number n . Is it possible to observe the quantization of energy levels of the ball?

Q.21. Show that:

- (a) $[A, B] + [B, A] = 0$
- (b) $[A, B + C] = [A, B] + [A, C]$

Q.22. Select the correct options:

i) Which compound would be expected to exhibit intense IR absorption band at 3300 cm^{-1} ?

a) Butane, b) 2-butene, c) $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$, d) But-1-ene

ii) $\text{C}\equiv\text{C}$ stretching frequency is less than $\text{C}\equiv\text{N}$ stretching frequency due to

- a) Higher dipole moment of $\text{C}\equiv\text{N}$ compared to $\text{C}\equiv\text{C}$
- b) Stronger force constant of $\text{C}\equiv\text{N}$ compared to $\text{C}\equiv\text{C}$
- c) Lower reduced mass of $\text{C}\equiv\text{N}$ compared to $\text{C}\equiv\text{C}$
- d) None of the above

iii) Ethyne ($\text{CH}\equiv\text{CH}$) does not show an IR absorption band in the region $2000\text{--}2200 \text{ cm}^{-1}$ because

- a) The stretching frequency for $\text{C}\equiv\text{C}$ comes around 1600 cm^{-1}
- b) There is no change in polarizability of the molecule due to vibration
- c) There is no change in dipole moment when ethyne $\text{C}\equiv\text{C}$ stretches
- d) The energy required for $\text{C}\equiv\text{C}$ stretching are lower

iv) Raman spectroscopy is not a resonance phenomenon because

- a) It measures scattering of photons
- b) There are no real transitions among energy levels
- c) No EMR energy matches with the process of Raman scattering
- d) There is no selection rule applicable for Raman spectroscopy

Q.23. Justify the following statements:

a) CO_2 has four normal modes of vibration however only two IR bands are observed.

b) Infinitely dilute solution of benzoic acid exhibits a sharp IR peak at 3600 cm^{-1} while neat benzoic acid exhibits a broad band 3400 cm^{-1} .

Q.24. Why is it advised to use longer wavelength laser for measurement of Raman scattering? State two differences between Fluorescence photons and Raman scattered photons.

Q.25. Water is a poor solvent for IR measurements of solution while water can be used as solvent for Raman measurement. Give two reasons for the statement. What is the mutual exclusion principle?